

Lab 10: Quantum Optics on the Bench

Single photons on a tabletop kit

1 ECTS · in person at i-LAMP

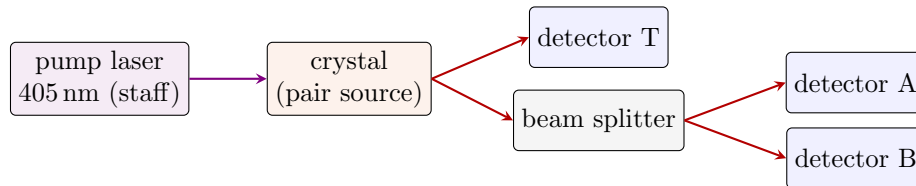
Prerequisites: *Fundamental Concepts of Quantum Technologies* (lecture series) — nothing else.

What you need: the Thorlabs EDU-QOP1 Quantum Optics Kit with its time tagger and software (set up by the staff); the worksheet (pages 4–5); a pen and a calculator.

Time: about 10 h, indicative: Part A 2 h, B 1.5 h, C 2 h, D 2.5 h, optional parts 2 h.

Source: Thorlabs, *EDU-QOP1(/M) Quantum Optics Kit User Guide*, Rev. D (2025-10-25), “manual §x”; follow its sections, this guide does not repeat them.

Goal: decide, from detector clicks alone, whether the light on the bench comes as single photons, and whether one photon can interfere with itself.



Parts B–D. A: dimmed 635 nm alignment laser into the beam splitter, no T. D: Michelson before B.

Safety first

The room is dark during measurements. The 405 nm pump laser is class 3B: **only the staff switch it on** (key, interlock; manual Ch. 1, Ch. 14). While it is on (Parts B–D) you follow the i-LAMP eyewear and conduct rules of the laser safety officer; the manual is explicit: “Always wear laser safety glasses when working with the pump laser!” (manual §7.6.4). You use the 635 nm alignment laser and the software.

Words you need

Beam splitter: a half-transparent mirror with two outputs (course document QC10, *Light*). **Click:** one pulse of a single-photon detector. **Coincidence:** two (or three) detectors click in the same **window** Δt (5 ns here, set in the software). **$g^{(2)}$:** with $P(X)$ the probability that detector X clicks in one window, $P(X) = R_X \Delta t$ for a click rate R_X (clicks per second),

$$g^{(2)} = \frac{P(A \text{ and } B)}{P(A)P(B)} : \quad 1 = \text{no correlation}, \quad < 1 = \text{fewer coincidences than chance}, \quad > 1 = \text{more}.$$

Accidentals: the coincidences expected by pure chance if A and B click independently, rate $R_A R_B \Delta t$; so $g^{(2)} = \text{coincidences} / \text{accidentals}$. **Poisson light:** photons arriving independently at random times, like raindrops; a laser, however dim, is Poisson light. **Pair source:** a crystal that, pumped at 405 nm, emits two photons (about 810 nm) at once. **Heralding:** a click at T announces that its partner photon is on the way to the beam splitter; A and B are counted only together with T. **Interference:** a wave split in two and recombined adds up or cancels, depending on the path difference; **visibility** $\mathcal{V} = (R_{\max} - R_{\min}) / (R_{\max} + R_{\min})$, 1 for perfect fringes. **Standard deviation** σ : the typical scatter of repeated results; a count of N clicks scatters by about $\sqrt{N}$.

Part A — Is a very dim laser a single-photon source?

A single photon cannot split: at the beam splitter it reaches A *or* B, never both, so a true single-photon source gives $g^{(2)} = 0$. Poisson light is different: each photon picks its exit on its own, so A and B click independently and $g^{(2)} = 1$ exactly, at *every* brightness. The alignment laser is dimmed by the kit’s filter; you look at A and B (Hanbury Brown–Twiss, HBT).

Formula → Number → Bench

$g_{\text{HBT}}^{(2)} = \frac{R_{AB}}{R_A R_B \Delta t}$. Manual §9.1, Table 1 ($\Delta t = 5$ ns): accidentals $R_A R_B \Delta t = 617.9$ Hz, measured $R_{AB} = 619.1$ Hz, so $g_{\text{HBT}}^{(2)} = 1.0020$ (the manual prints 1.00201).

Predict → measure → compare (worksheet W1)

Predict $g^{(2)}$. **Measure** following manual §9.1 (HBT tab, §11.7): its 20 s measurement, three times at the kit's setting. **Compare** with 1, with Table 1 (1.0020), and run to run. Dimmer light only enlarges the error bar (manual fn. 84): try it in the pre-lab playbox.

Counterintuitive — and Real

Manual §9.1 estimates how many photons are inside the 1 m setup at any moment: $n = lf/c = 1 \text{ m} \times 1.2 \cdot 10^6 \text{ s}^{-1} / 3 \cdot 10^8 \text{ m s}^{-1} = \mathbf{0.004}$ photons. Far below one — surely single photons? Yet $g^{(2)} \approx 1$ (Table 1: 1.0020), not 0. Rare photons are not single photons: a dim laser stays Poisson.

Part B — Does the crystal really make pairs?

The staff switch on the pump. If the crystal makes pairs, T and A click together far more often than by chance.

Formula → Number → Bench

$g_{\text{PS}}^{(2)} = \frac{R_{TA}}{R_T R_A \Delta t}$ = coincidences / accidentals. Manual §9.4, Table 2 (rates = counts / 20 s): $R_T = 231900 \text{ Hz}$, $R_A = 145250 \text{ Hz}$, $R_{TA} = 19293.25 \text{ Hz}$, so $g_{\text{PS}}^{(2)} \approx 115$: coincidences are $\approx 115 \times$ the accidentals. Table 2 prints no Δt ; we assume the standard $\Delta t = 5 \text{ ns}$ (manual §7.4.5).

Predict → measure → compare (worksheet W2)

Predict: the manual expects $g_{\text{PS}}^{(2)} \geq 30$ (§9.2). **Measure** following manual §9.2: Alignment tab (manual §9.2 calls it Adjustment; §11.5) or GRA tab (§11.7), detectors T and A. **Compare** with 30 and with Table 2.

Part C — Can one photon take both exits?

Now the partner photon, heralded by T, meets the beam splitter (Grangier–Roger–Aspect, GRA). Any classical light obeys $g^{(2)} \geq 1$ (manual eq. 75); one photon gives 0. The software counts a triple as a *coincidence of coincidences*: T&A and T&B both coincident (manual §4.2.2, §12.2.1).

Formula → Number → Bench

$g_{\text{GRA}}^{(2)} = \frac{N_{TAB} N_T}{N_{TA} N_{TB}}$ (counts: the measurement time cancels). Manual §9.4, Table 2 (20 s): $N_T = 4.638 \text{ M}$, $N_{TA} = 385865$, $N_{TB} = 324688$, $N_{TAB} = 513$, so $g_{\text{GRA}}^{(2)} = 0.0190$. With $\sigma \approx g_{\text{GRA}}^{(2)} / \sqrt{N_{TAB}} = 0.00084$, it lies $(1 - g_{\text{GRA}}^{(2)}) / \sigma \approx 1170$ standard deviations below the classical bound.

Predict → measure → compare (worksheet W3)

Predict: $g_{\text{GRA}}^{(2)} < 1$. **Measure** following manual §9.4 (GRA tab, §11.7): four counts and the time. **Compare:** how many σ below 1? Is the residual chance (next box)?

The bench is not the model: the window Δt

Detectors and time tagger resolve time only to about a nanosecond, so coincidences are counted in a window, and unrelated clicks in the same window count too. Two chance types fake triples (manual eqs. 105–106): $R^{(3)} = R_T R_A R_B \Delta t^2$ and $R^{(2+1)} = (R_{TA} R_B + R_{TB} R_A) \Delta t$, giving $g_{\text{acc}}^{(2)} = (R^{(3)} + R^{(2+1)}) R_T / (R_{TA} R_{TB})$. Table 2: $g_{\text{acc}}^{(2)} = 0.0193$, that is ≈ 520 chance triples in 20 s against 513 measured, a difference of 0.33σ . **The residual $g_{\text{GRA}}^{(2)}$ is consistent with chance coincidences alone: no sign of the photon splitting.** A shorter window or weaker pump lowers it (eq. 110).

The bench is not the model: real detectors

Our model is idealized. Detector efficiency cancels in every $g^{(2)}$, except that of the herald T in $g_{\text{GRA}}^{(2)}$: a T that misses photons raises its small two-pair term by up to a factor 2 (it stays far below 1). Dark counts and stray light enter only as accidentals.

Part D — Does a single photon interfere with itself?

The staff place a Michelson interferometer in one beam-splitter output, with B behind it: a second beam splitter sends the photon into two arms, mirrors send it back, and the arms recombine. The stage moves one mirror by x ; the light goes out and back, so the path difference is $2x$. The detector-side probability is $P_d(x) = (1 - \cos(4\pi x/\lambda))/2$, so R_{TB} rises and falls with period $\lambda/2$ in mirror position, while $g_{\text{GRA}}^{(2)} < 1$ along the scan.

Formula → Number → Bench

With $\lambda \approx 810$ nm the fringe period is ≈ 405 nm in mirror position (manual §11.9: “about 400 nm”). Backwards, one maximum per fringe: $\lambda = 2 \times (\text{mean spacing of maxima}) = 2(x_{\text{last}} - x_{\text{first}})/(N_{\text{max}} - 1) \approx 810$ nm, with a stage error of up to 3 % (manual §9.7).

Predict → measure → compare (worksheet W4)

Predict the period and λ . **Measure** following manual §9.7 (Michelson tab, §11.9); read $g_{\text{GRA}}^{(2)}$ during the scan. **Compare** λ with $810 \text{ nm} \pm 3\%$, and $\mathcal{V}$ with 1. Is $g_{\text{GRA}}^{(2)} < 1$ at every position? Then the fringes are built photon by photon.

Optional (not needed for the core): Parts E, F and two controls

Polarization is the direction in which the light’s field oscillates; a **polarizer** passes one direction; a **half-wave plate** mirrors the direction about its axis. The pair photons are vertically polarized. **E, quantum eraser** (manual §7.7, §9.8): a half-wave plate at 22.5° between the two beam splitters makes the polarization diagonal. Arm polarizers crossed at 0° and 90° mark the path: no fringes, $P_d = \frac{1}{4}$ flat. A polarizer at $\pm 45^\circ$ before the detector erases the mark and the fringes return, $P_d = (1 \pm \cos \varphi)/8$ with $\varphi = 4\pi x/\lambda$; with both arm polarizers at 0° the maximum is twice the flat level. **F, Malus’s law for single photons** (manual §9.6): a polarizer at θ from the horizontal passes a heralded photon with probability $\sin^2 \theta$. **Controls.** Manual §9.3, one arm alone (T blocked): *thermal light* (many independent emitters, whose photons bunch) would give 2, but its coherence time (≈ 0.2 ps) is far below Δt , so expect ≈ 1 , not 2. Manual §9.5, GRA with classical light: expect ≈ 1 ; three detectors alone do not make $g^{(2)} < 1$.

What this kit cannot show: entanglement

The course plan promises entangled photons. This kit’s single crystal makes pairs with both photons vertically polarized, $|VV\rangle = |V\rangle \otimes |V\rangle$: “correlated in time and energy (though not polarization-entangled)” (manual Ch. 2). *Entangled* means a two-photon state that cannot be written as $|\alpha\rangle \otimes |\beta\rangle$, such as $|\Phi^+\rangle = (|HH\rangle + |VV\rangle)/\sqrt{2}$ (H : horizontal). The optional notebook section compares the two with a polarizer on each photon: at the standard angles the CHSH sum S (a combination of four polarizer correlations, defined in the optional notebook section) is $S = \sqrt{2} \approx 1.41$ for $|VV\rangle$ against $2\sqrt{2} \approx 2.83$ for $|\Phi^+\rangle$, while every *local hidden variable model* (each photon leaves the source with ready-made answers for every polarizer angle) obeys $|S| \leq 2$. The Thorlabs *EDU-QOPA1(/M) Polarization-Entanglement Extension Kit* adds a crossed type-I BBO pair plus compensation crystals; it is not used here. *Anchor*: the Micius satellite sent entangled pairs to two ground stations (2017).

Real-World Anchors

A: Hanbury Brown and Twiss (1956) correlated two detectors’ signals, the method behind $g_{\text{HBT}}^{(2)}$; it measured the sizes of stars. **C:** Grangier, Roger and Aspect (1986) ran Part C with heralded photons from atoms and found 0.18; today Quandela builds quantum-dot single-photon sources for photonic quantum computers. **D:** LIGO’s kilometre-long Michelson interferometers reported gravitational waves (2016). **Optional:** Nobel Prize in Physics 2022 (Aspect, Clauser, Zeilinger), entangled photons.

References. Thorlabs, *EDU-QOPA1(/M) Quantum Optics Kit User Guide*, Rev. D (2025-10-25): Ch. 1, 2, 14; §4.2.2, §7.6.4, §7.7, §9.1–9.8, §11.7, §11.9, §12.2.1. Course document QC10, *Light*. R. Hanbury Brown, R. Q. Twiss, *Nature* **177**, 27 (1956). P. Grangier, G. Roger, A. Aspect, *Europhys. Lett.* **1**, 173 (1986). J. F. Clauser, M. A. Horne, A. Shimony, R. A. Holt, *Phys. Rev. Lett.* **23**, 880 (1969). Nobel Prize in Physics 2022 (Aspect, Clauser, Zeilinger). J. Yin et al., *Science* **356**, 1140 (2017). B. P. Abbott et al., *Phys. Rev. Lett.* **116**, 061102 (2016). N. Somaschi et al., *Nat. Photon.* **10**, 340 (2016).

Worksheet — Lab 10: Quantum Optics on the Bench

Group _____ Names _____ Date _____

W1 · Part A: attenuated laser (HBT), kit setting, 20 s per run. $\Delta t = \rule{1cm}{0.4pt}$ ns. Rates in Hz; accidentals = $R_A R_B \Delta t$; $g_{\text{HBT}}^{(2)} = R_{AB} / \text{accidentals}$.

run	R_A	R_B	R_{AB}	accidentals	$g_{\text{HBT}}^{(2)}$	predicted
run 1						1
run 2						1
run 3						1
<i>Optional:</i> extra ND filter (only if i-LAMP has one; tens of minutes per run)						
extra ND filter						1

Compare: Table 1 gives 1.0020. Your runs: consistent with 1 and with each other? _____

W2 · Part B: pair source. $g_{\text{PS}}^{(2)} = R_{TA} / (R_T R_A \Delta t)$.

	R_T	R_A	R_{TA}	$R_T R_A \Delta t$	$g_{\text{PS}}^{(2)}$	predicted
your run						≥ 30

Compare: at least 30? Table 2 gives ≈ 115 . _____

W3 · Part C: heralded photons (GRA). $g_{\text{GRA}}^{(2)} = N_{TAB} N_T / (N_{TA} N_{TB})$; $\sigma \approx g_{\text{GRA}}^{(2)} / \sqrt{N_{TAB}}$. The Table 2 row is the manual's example.

	time (s)	N_T	N_{TA}	N_{TB}	N_{TAB}	$g_{\text{GRA}}^{(2)}$	σ	$(1 - g^{(2)}) / \sigma$
Table 2	20	4.638 M	385865	324688	513	0.0190	0.00084	1170
your run								
predicted	one photon cannot take both exits					< 1		

Accidentals (the window box). Rates $R_X = N_X / \text{time}$ in Hz; $R^{(3)} = R_T R_A R_B \Delta t^2$; $R^{(2+1)} = (R_{TA} R_B + R_{TB} R_A) \Delta t$; $g_{\text{acc}}^{(2)} = (R^{(3)} + R^{(2+1)}) R_T / (R_{TA} R_{TB})$; chance triples = $(R^{(3)} + R^{(2+1)}) \times \text{time}$.

	R_T	R_A	R_B	R_{TA}	R_{TB}
Table 2	231900	145250	146250	19293.25	16234.4
your run					
	$R^{(3)}$ (Hz)	$R^{(2+1)}$ (Hz)	$g_{\text{acc}}^{(2)}$	chance triples	$\frac{\text{chance} - N_{TAB}}{\sqrt{N_{TAB}}}$
Table 2	0.1232	25.898	0.0193	520	0.33
your run					

Compare: is $g_{\text{GRA}}^{(2)}$ below 1 by many σ ? Do chance triples explain your N_{TAB} ? _____

W4 · Part D: heralded Michelson. Positions x of the fringe maxima of R_{TB} (in μm), in order; one maximum per fringe. Record at least the first and the last, and count them all for N_{max} .

maximum	1	2	3	4	5	6	7	8	9	10
x (μm)										
maximum	11	12	13	14	15	16	17	18	19	20
x (μm)										

	N_{max}	x_{first}	x_{last}	spacing (nm)	λ (nm)	R_{max}	R_{min}	$\mathcal{V}$
your scan								
predicted				≈ 405	≈ 810			1 (ideal)

$\lambda = 2(x_{\text{last}} - x_{\text{first}})/(N_{\text{max}} - 1)$. $g_{\text{GRA}}^{(2)}$ at a maximum _____, a minimum _____, in between _____ (predicted: all < 1).

Compare: is λ within 3% of 810 nm? _____

W5 · Optional parts.

F: polarizer θ	0°	30°	45°	60°	90°	
predicted $\sin^2 \theta$	0	$\frac{1}{4}$	$\frac{1}{2}$	$\frac{3}{4}$	1	
measured (relative)						

E: eraser	arms crossed	crossed + $\pm 45^\circ$	both arms 0°
predicted	no fringes	fringes	fringes
observed			

controls	one arm alone (§9.3)	classical GRA (§9.5)
predicted $g^{(2)}$	≈ 1	≈ 1
measured $g^{(2)}$		